

Retail Radar

Jul 15 — Trading Earnings as Flows Cool

Retail Trading Activity Overview

- **Retail flows decelerated to 24% below their YTD average**, driven by weaker ETF flows (18%ile) while single-stock activity also slowed (to 51%ile from 86%ile last week). Attention began to shift toward the 2Q earnings season, which kicked off Tuesday with strong bank results.
- **ETF weakness was broad-based.** The largest segment, large-cap ETFs, saw subdued inflows this week (-1.0z). Within sectors, positioning in **Semis** turned modestly more defensive, with investors selling SOXL (-1.8z) and buying SOXS (1.6z). Activity in **energy** and crude-linked ETFs was muted (BNO -0.4z; USO -0.3z; XLE 0.0z; SCO 0.2z). By contrast, **financials ETFs stood out, posting their largest inflows in 14 months, 2.5z**, Figure 1.
- Retail investors continue to add exposure to Tech and Communications. The **Mag 7** have reasserted themselves as positioning in Semis has softened so far in July: NVDA, TSLA, and MSFT lead month-to-date buying, alongside SPCX and SNDK (with SNDK flows plateauing since July 2, Figure 2).
- **The 2Q Earnings season opened on a strong note:**
 - **Banks** delivered robust results powered by their trading and IB divisions, see Macro & Fundamental Stories. Retail activity was mixed after the announcements and largely tracked performance: investors took profits in GS (-2.1z) and BAC (-0.4z) on Tuesday, while trimming WFC (0.5z).
 - **Airlines** also reported solid results, absorbing higher fuel expenses and supported by resilient travel demand, disciplined capacity growth, and continued strength in higher-margin revenue streams. Retail investors bought UAL (0.4z) ahead of their beat on Wednesday.
 - Meanwhile, in Tech, **ASML** reported strong earnings driven by surging AI demand: the company raised its 2026 full-year sales guidance and announced plans to boost production capacity by 30%. Retail flows were net positive in the name, with investors buying 1.0z on Wednesday and 1.9z over the week.
 - **IBM** ended trading Tuesday down more than 25%, its **worst drop since at least 1968**, after the company preannounced earnings that fell well below expectations. IBM quickly rose to the **most heavily sold stock among retail investors this week** (-6.0z week), following two consecutive -7.0z days on Tuesday and Wednesday. According to our analysts, the company cited the following as the main drivers of the underperformance, [report](#): 1) Shift in quarterly capex toward servers, storage, and memory in the last few weeks of June to secure supply-constrained infrastructure ahead of anticipated price increases, leading to weaker Z Systems performance. 2) Large deals failing to close, which drove the majority of the shortfall. 3) Transaction Processing weakness, which our analysts expect is due to lower up-front MIPS-driven consumption.
 - **Looking ahead, our analysts remain constructive Semis and Semis Equipment**, one of the most closely followed themes among retail investors, [report](#). Our analysts view the ~15% pullback in stocks as a

Equity Strategy and Quantitative Research

Arun Jain ^{AC}

(1-212) 622-9454
arun.p.jain@jpmorgan.com
J.P. Morgan Securities LLC

Shizuka Suga, CFA ^{AC}

(1-212) 622-2134
shizuka.suga@jpmorgan.com
J.P. Morgan Securities LLC

Ana Pous Avila

(1-212) 622-0496
ana.pousavila@jpmorgan.com
J.P. Morgan Securities LLC

William Matheson

(1-212) 622-9538
william.matheson@jpmchase.com
J.P. Morgan Securities LLC

Khuram Chaudhry

(44-20) 7134-6297
khuram.chaudhry@jpmorgan.com
J.P. Morgan Securities plc

Bhupinder Singh ^{AC}

(1-212) 622-9812
bhupinder.singh@jpmorgan.com
J.P. Morgan Securities LLC

Dubravko Lakos-Bujas ^{AC}

(1-212) 622-3601
dubravko.lakos-bujas@jpmorgan.com
J.P. Morgan Securities LLC

Past Issues:

[7/8 - Split-the-Dip as Momentum Cools](#)
[7/1 - Same Playbook, Buying the Dip](#)
[6/24 - Remembers the Dip, MU Serves Right](#)
[6/17 - Tech Craze Spillover to Communications](#)
[6/10 - Tech Selling amidst Drawdown, World Cup](#)
[6/3 - MEME Earnings, Trading around IPOs](#)
[5/27 - Theme Stack: AI, Memory, Space, Quantum](#)
[5/13 - Software vs Semis Positioning](#)
[5/6 - Memory and AI, the Catch-Up Trade](#)
[4/29 - Peak Earnings, Peak Mag-7](#)
[4/22 - Towards a MEME Revival?](#)
[4/15 - Back to Average](#)
[4/8 - Not Buying Despite the Ceasefire](#)
[4/1 - In Defensive Mode](#)
[3/25 - Selling Rips](#)
[3/18 - From FOMO to FOHO: Fear Of Higher Oil](#)
[3/11 - Buying Oil, Selling Energy](#)
[3/4 - Cautiously Optimistic](#)
[2/25 - Balanced Optimism](#)

springboard for 2H outperformance and expect the upcoming positive earnings revision cycle to continue, supported by AI-led momentum and a broadening cyclical recovery. Retail flows in MU (+0.3z) and SNDK (-0.7z) were moderate, consistent with the broader plateau in retail activity across these names in July.

[2/18 - AI Software Averaging Down](#)
[2/11 - Positioning around Dislocations](#)
[2/4 - Softening Sentiment, Software vs Semis](#)
[1/28 - Earnings, Intl Equities, Precious Metals](#)
[1/21 - Unfazed through Vol Shock](#)
[1/14 - The January Effect?](#)
[1/7 - December Spree and a good start to January](#)
[12/10 - 2025 Retail Mania](#)
[12/3 - The Retail AI Trade](#)
[11/26 - Weekly Highlights](#)
[11/19 - 2025 Retail Mania: Dip-buying in AI](#)
[11/12 - Bullion to Silicon: AI Gold Rush Resumes](#)
[11/5 - Skipping the Dip, Earnings Stock Picking](#)
[10/29 - Reaching Peak-Earnings, AI, FOMC](#)
[10/22 - Sold GLD, onto New Memes and Earnings](#)
[10/15 - Trading around Tariffs, AI and Earnings](#)
[10/8 - Back after Summer Break](#)
[10/1 - Back to Stocks, Thematic ETFs, GLD Rush](#)
[9/24 - Precious Metals, Cyclical and Domestic](#)
[9/17 - FOMC, Herding in OPEN and GRAB](#)

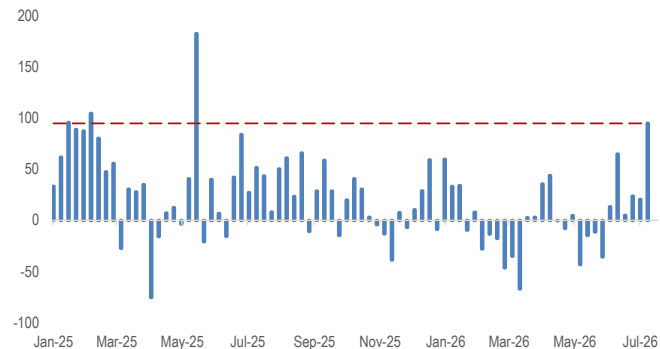
We thank G. K. Kiruthik Srinivaas and Aditya Mulye of J.P. Morgan India Private Limited for their significant contributions to this report.

Retail Trading Activity in Numbers for the Week: Jul 9 to Jul 15

- Retail flows were \$5.6B this week, below the 12-month avg of \$6.8B/week. Retail investors continued to favor ETFs (+\$4.6B) over Single Stocks (+\$1.0B).
- Outside of **Broad Based Large Cap Equity ETFs** (+\$1.2B, -1.0z), retail ETF buying was influenced by **Fixed Income** — **Multi Sector** (+\$210M), **Equity Style Dividend** (+\$200M), **Equity Style Call/put writing** (+189M), and **Equity Broad Based Multi Cap** (+\$161M).
- This week, in line with prior weeks, retail investors **continued to buy AI datacenters and electrification (JPAMAIDE), Top 30 AI/Datacenter Beneficiaries, Mag 7, Growth**, along with **US Companies with High Direct China Exposure** and **OBBBA Immediate Expensing Beneficiaries**, see Figure 4.
- **Activity in Mag7** this week: Retail investors bought NVDA (+\$486M), TSLA (+\$343M), GOOGL/GOOG (+\$182M), MSFT (+\$167M), AMZN (+\$54M), META (+\$2M) and AAPL (-\$120M).
- **Outside of Mag7**, retail investors favored Tech (+\$362M) and Communications (+\$267M) and were net sellers of Financials (-\$306M) and Cons. Disc. (-\$231M).
- **Top 5 retail stocks last week:** NVDA (+\$486M), SPCX (+\$351M), TSLA (+\$343M), GOOGL (+\$174M) and INTC (+\$171M). **Bottom 5 retail stocks:** IBM (-\$202M), AAPL (-\$120M), AAL (-\$61M), CRM (-\$60M) and PANW (-\$49M).
- **In options**, retail share is at highs, Figure 49. The most traded options were in the following names, echoing previous weeks: TSLA, NVDA, MU, AMZN, SNDK, AMD, GOOGL/GOOG, INTC, ORCL and NBIS. For the most bought/sold delta and gamma names, see Retail Activity in Options. Also see Retail Investors Options Activity by Sectors.
- **Futures Activity** (Non Retail): Over the past week, Futures traders net sold ~\$0.6B, primarily driven by net sales in ES (~\$7.5B) and partially offset by net buys in RTY (~\$8.1B).

Figure 1: Retail Imbalance in Financials at their 14m highs

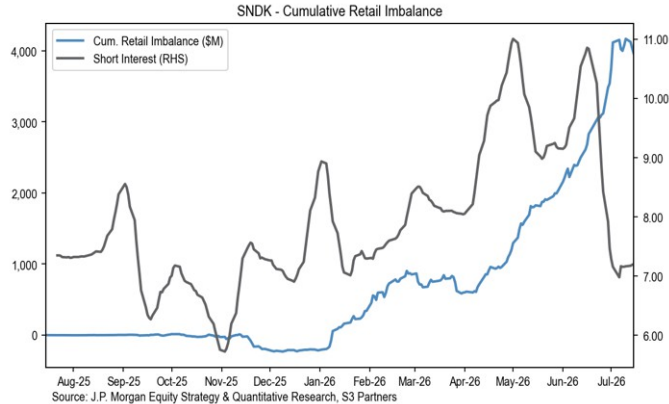
As of July 15th



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 2: Cumulative Retail Imbalance in SNDK

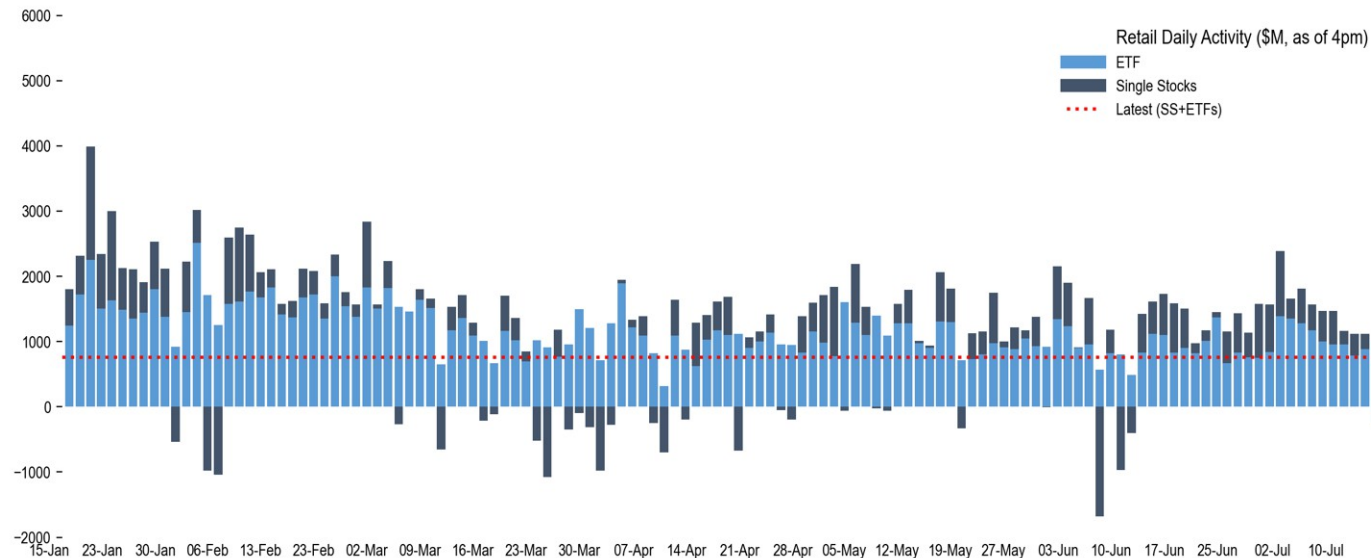
As of July 15th



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 3: Retail Investor Daily Purchases by Stocks and ETFs

\$M, as of Jul 15th



Source: J.P. Morgan Equity Strategy & Quantitative Research

Retail Cash Activity – Overview

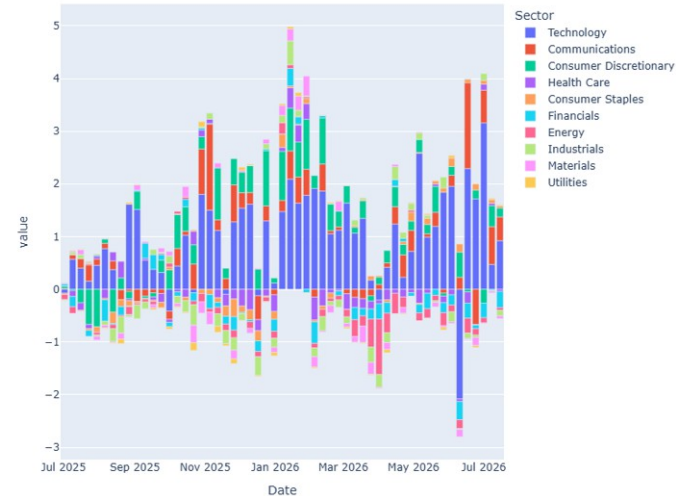
Figure 4: Retail Single Stock Activity by Themes

Thematic Baskets		Δ in Retail Turnover (\$Bn)		
Basket	Index	Week Ago	Month Ago	YTD
AI/Datacenter/Electrification	JPMAIDE	1.27	7.28	43.40
S&P 500 Top 30 AI/Datacenter Beneficiaries	JPRAID30	1.07	7.13	42.38
Magnificent 7	Mag7	0.97	4.86	34.22
Growth	JPAMGROW	0.96	6.28	41.29
US Companies with High Direct China Revenue Exposure	JPAMCREV	0.94	3.45	22.28
OBDBA Immediate Expensing Beneficiaries	JPGTIEXB	0.61	4.26	20.37
Sensitive to Low-End Consumer	JPRLOWCO	-0.02	-0.16	-1.47
Geopolitically Sensitive Energy Companies	JPAMGNRG	-0.02	-0.13	-1.95
US Federal Budget Sensitive	JPAMGOVT	-0.03	-0.18	0.23
US Inflation	JPAMINFL	-0.03	-0.24	-1.98
Trump Deregulation Agenda	JPRGDREG	-0.07	-0.30	-4.19
Inflation Outperformers	JPAMINOP	-0.08	-0.63	-3.44

Source: J.P. Morgan Equity Strategy & Quantitative Research. See [report](#), [report](#), and [report](#) for details on above baskets. Note: J.P. Morgan Research does not provide research coverage of the baskets mentioned here, and investors should not expect continuous analysis or additional reports relating to them.

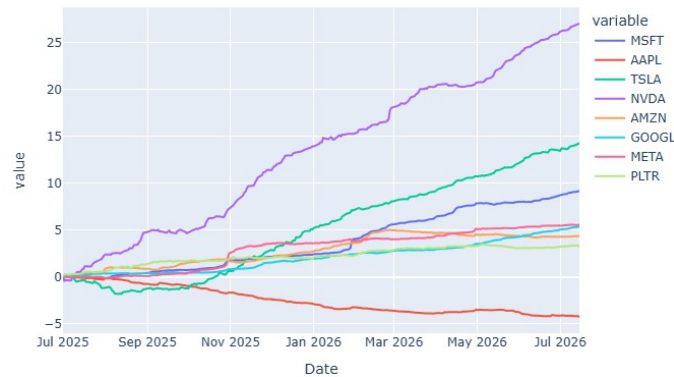
Figure 6: Retail Single Stock Activity by Sector (\$B)

Activity is dominated by Tech and Cons. Disc.



Source: J.P. Morgan Equity Strategy & Quantitative Research

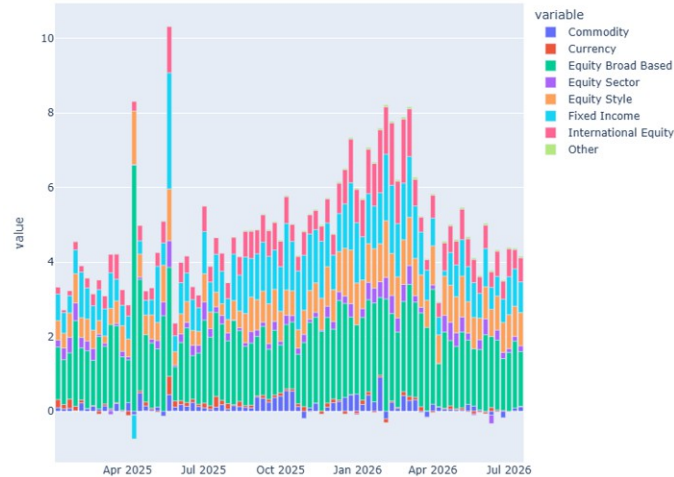
Figure 5: Retail Cumulative Purchases in Mag 7 + PLTR (\$B)



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 7: Retail ETF Activity by Themes

Buying / Selling of ETFs aggregated by themes, in \$B.



Source: J.P. Morgan Equity Strategy & Quantitative Research

Single Stock Stories with Elevated Retail Activity

Hedge Funds vs. Retail: High Short Interest Meme Stocks

- **EOSE (\$3.6M bought last week, 0.5z, Figure 10):** The company launched a \$150M rights offering, with proceeds earmarked for an investment in Frontier Power USA, alongside a \$75M equity investment from Hudson Bay Capital. Retail interest has re-emerged since January 2026, with net purchases totaling \$40M, amid elevated short interest of ~33%, up 7% over the same period.
- **IREN (\$25.6M bought last week, 0.5z, Figure 11):** The company has recently undergone management changes, and shares fell ~12% earlier this month amid governance-related backlash over compensation and broader Nasdaq weakness. Retail buying has accelerated since September 2025, with net purchases totaling \$400M, alongside high short interest of ~24.5% of float, which has increased by 10% since May 2026.
- **UNCY (\$0.6M sold last week, -1.5z, Figure 12):** The stock dropped as much as 40% on June 30, 2026 after the FDA issued a second complete response letter indicating the NDA for oxylanthanum carbonate was not approved. Retail investors have been actively buying since May 2026, with net purchases totaling \$7.5M, alongside moderate short interest of ~20% of float, up 10% over the same period.

Earnings Stories

2Q26 earnings season is underway. As of 7/15, 7.2% of S&P 500 companies (~9% of total market cap) have reported, with results concentrated in Financials and Industrials. Consensus expects 2Q26 earnings to grow +23.7% y/y (vs. a +18.1% avg. last 4Qs), driven primarily by Tech and Energy. Revenue growth is forecast at +11.7% y/y (vs. +8.8% avg. last 4Qs). All sectors are expected to post positive revenue growth, and all sectors except Healthcare are projected to deliver positive earnings growth.

- **Banks** reported robust earnings, with [Goldman Sachs](#) (GS, \$20.8M sold on 7/14, -3.5z, Figure 13), [Bank of America](#) (BAC, \$12.1M sold on 7/14, -1.4z, Figure 15), [Wells Fargo](#) (WFC, \$7.7M sold on 7/14, -1.3z, Figure 17), and [Citigroup](#) (C, \$11.4M bought on 7/14, 3.0z, Figure 19) all beating market expectations. Results were broadly supported by strong trading and investment banking revenues, while management teams also pointed to solid loan and deposit growth, continued share repurchase activity, stable capital ratios, recovering wealth management fees, and resilient credit metrics. Stock performance was more mixed following the announcements, with GS and BAC up approximately +3–9%, while WFC and C declined roughly 3–5%.
- **Airlines** reported broadly solid 2Q26 results, with [Delta](#) (DAL, \$3.6M sold on 7/10, -1.2z, Figure 21) and **United** (UAL, \$3.6M bought on 7/15, 1.4z, Figure 23) both benefiting from resilient travel demand, disciplined capacity growth, and continued strength in higher-margin revenue streams. Delta stood out for strong premium, loyalty, and corporate demand, which helped support margins despite ongoing fuel cost pressure. United has also pointed to healthy demand and capacity discipline, though their outlook is more sensitive to fuel costs.
- **ASML's** (\$26.6M bought on 7/15, 1.1z, Figure 25) 2Q results and 3Q guidance were better-than-expected, with the company raising FY26 sales guidance for FY26 by 16%. Most importantly, the company indicated that they will increase both EUV and immersion DUV capacity in both 2027 and 2028 by 30%, which is higher than all current sell-side estimates. To this end, upcoming earnings, guidance and revisions for [Semiconductors/Semi Cap Equipment](#) in the US are also expected to be strong, which could support outperformance of the space through the rest of the year, especially after the recent 13-15% pullback. The sector remains amongst the most traded among retail investors (SOXS \$80M bought in the last week, Figure 27) with our analysts continuing to favor names with the highest-quality AI exposure and/or clean leverage to the broadening cyclical recovery, including **AVGO** (\$56.6M bought in the last week, 0.03z, Figure 29), **MRVL** (\$20.5M bought in the last week, 0.4z, Figure 31), **NVDA** (\$563.4M bought in the last week, 0.1z, Figure 33), **ADI** (\$3.3M bought in the last week, 0.2z, Figure 35), **MU** (\$133M bought in the last week, 0.1z, Figure 37), **KLAC** (\$1.6M bought in the last week, 0.2z, Figure 39), and **SNPS** (\$6.4M sold in the last week, -0.1z, Figure 41).
- **IBM** (\$95M bought on 7/14, -7.8z, Figure 43) negatively pre-announced 2Q results prompting a -25% sell-off. The company cited the following as main drivers of the underperformance: 1) Shift in quarterly capex toward servers, storage, and memory in the last few weeks of June to secure supply-constrained infrastructure ahead of anticipated price increases, leading to weaker Z Systems performance; 2) Large deals failing to close, which drove the majority of the shortfall; and 3) Transaction Processing weakness.

Meme Battlegrounds: Retail Buying vs. Hedge Fund Shorting

In this section, we are flagging a few stocks that are heavily mentioned on Social Media and have high Retail buying as well as high Hedge fund Shorting. These stocks may experience unexpected flows in case of increased activity.

Social Media Screen and Retail Activity

Figure 8: Most Hyped Stocks on Social Media

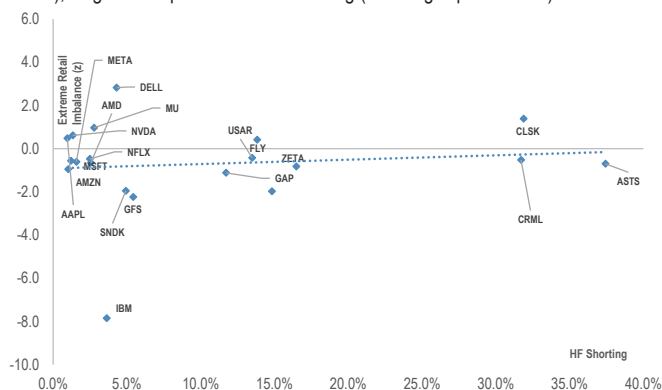
Top 20 names based on social media traction over the week (Social Interest), Scaled by market cap

Ticker	Name	Mkt Cap (\$B)	Social Interest Score	Median Short Interest (% float)	Extreme Imbalance 1D Z	Date of Imbalance
MU	Micron Technology Inc	1110.3	32.5	2.8%	1.0	09-Jul
META	Meta Platforms Inc	1678.0	31.3	1.6%	-0.6	13-Jul
MSFT	Microsoft Corp	2859.4	31.2	1.2%	-0.6	09-Jul
AMZN	Amazon.com Inc	2662.3	30.1	1.0%	-0.9	09-Jul
AAPL	Apple Inc	4624.5	30.1	1.0%	0.5	14-Jul
CLSK	Cleantech Inc	3.5	30.0	31.9%	1.4	14-Jul
NVDA	NVIDIA Corp	5125.6	29.4	1.3%	0.6	13-Jul
FLY	Firefly Aerospace Inc	3.5	29.0	13.8%	0.4	08-Jul
ZETA	Zeta Global Holdings Corp	5.6	28.7	16.5%	-0.8	14-Jul
NFLX	Netflix Inc	309.6	28.4	2.5%	-0.5	08-Jul
GAP	Gap Inc/The	7.2	28.4	11.7%	-1.1	08-Jul
CRML	Critical Metals Corp	1.2	28.4	31.7%	-0.5	13-Jul
GFS	GLOBALFOUNDRIES Inc	34.8	28.3	5.4%	-2.2	14-Jul
IBM	International Business Machines Corp	204.0	28.1	3.6%	-7.8	14-Jul
SNDK	Sandisk Corp	260.3	27.7	4.9%	-2.0	14-Jul
AMD	Advanced Micro Devices Inc	893.8	27.5	2.6%	-0.6	09-Jul
ASTS	AST SpaceMobile Inc	26.7	27.4	37.4%	-0.7	14-Jul
DELL	Dell Technologies Inc	295.6	27.4	4.3%	2.8	14-Jul
USAR	USA Rare Earth Inc	4.2	26.8	13.5%	-0.4	09-Jul
SOFI	SoFi Technologies Inc	23.8	26.7	14.8%	-2.0	09-Jul

Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

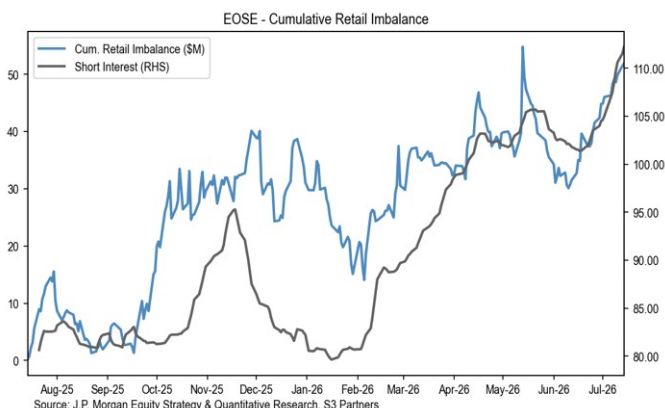
Figure 9: Increasing Squeeze Risk in Meme Stocks: Retail Buying in High Short Interest Names

Positive Slope = Opposite Positioning (Heightened Risk of Short Squeeze or Retail Losses); Negative Slope = Similar Positioning (Diffusing Squeeze Risk)



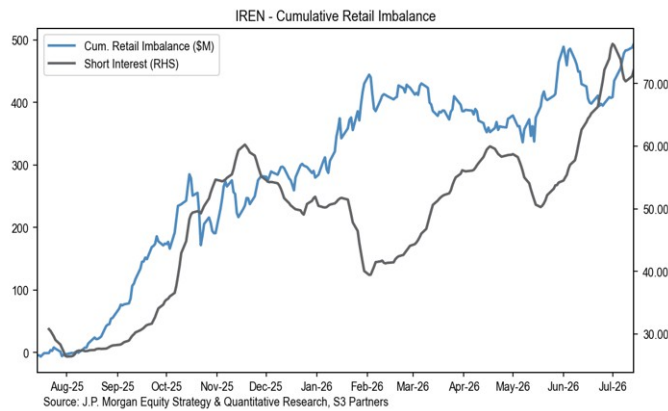
Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 10: Retail Activity in EOSE



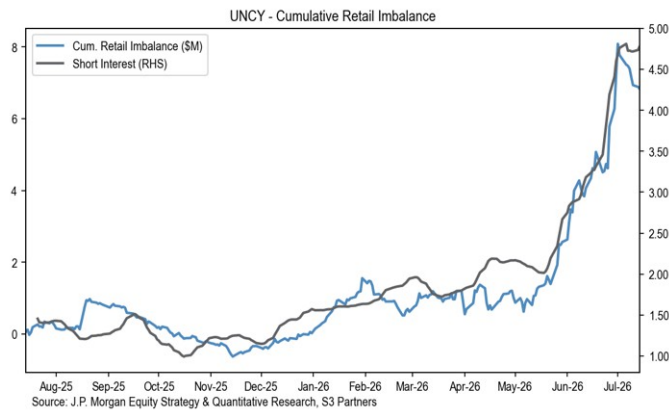
Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 11: Retail Activity in IREN



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 12: Retail Activity in UNCY

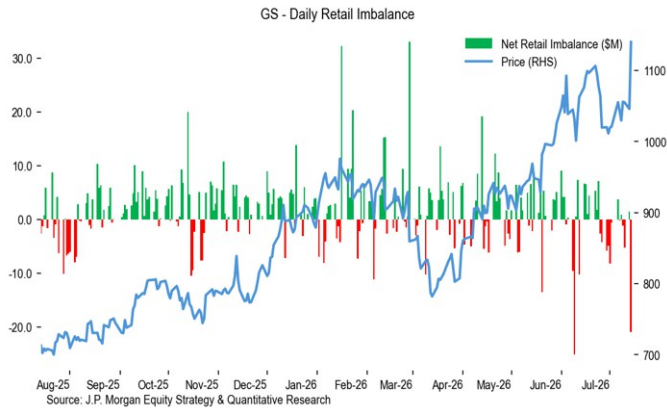


Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Macro & Fundamental Stories

Figure 13: Retail Imbalance in GS

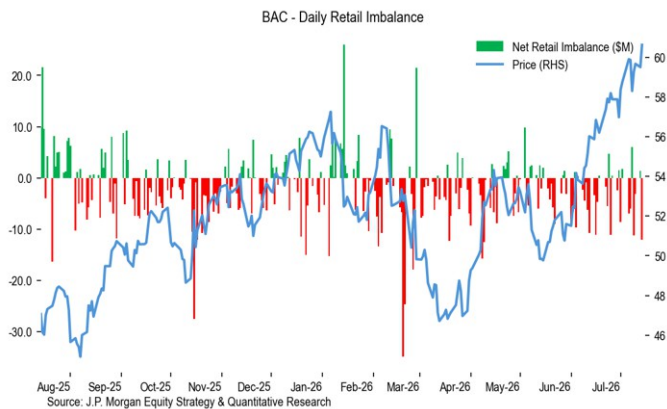
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 15: Retail Imbalance in BAC

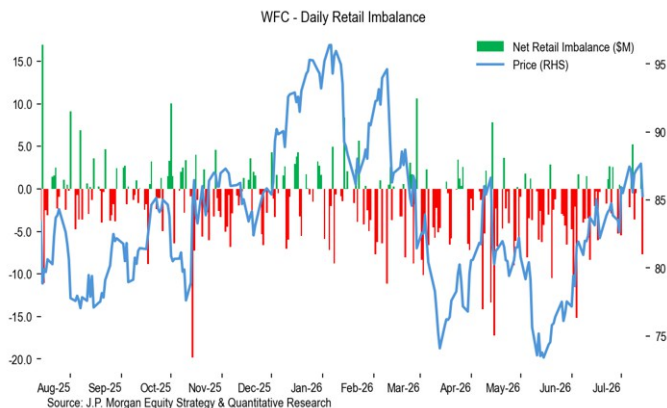
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 17: Retail Imbalance in WFC

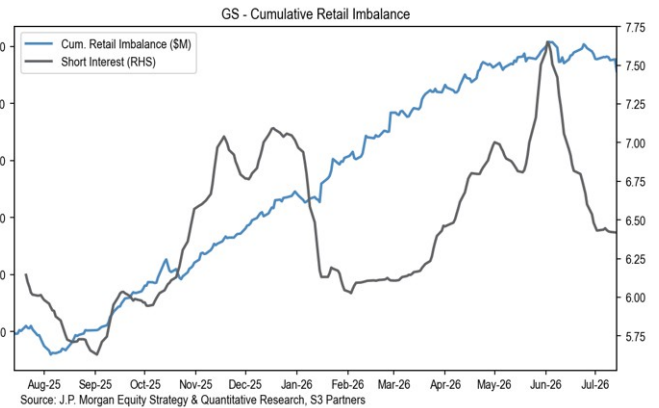
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 14: Cumulative Retail Imbalance in GS

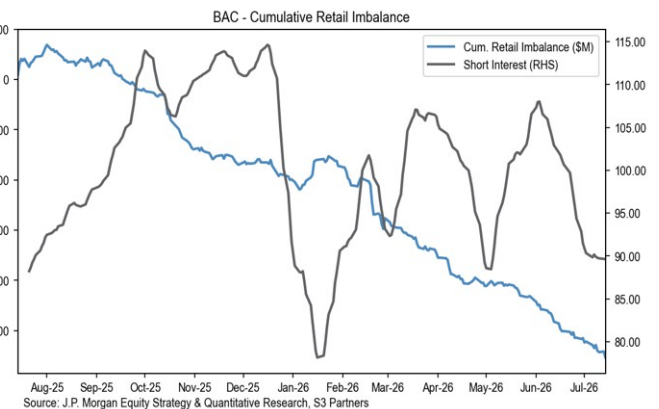
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 16: Cumulative Retail Imbalance in BAC

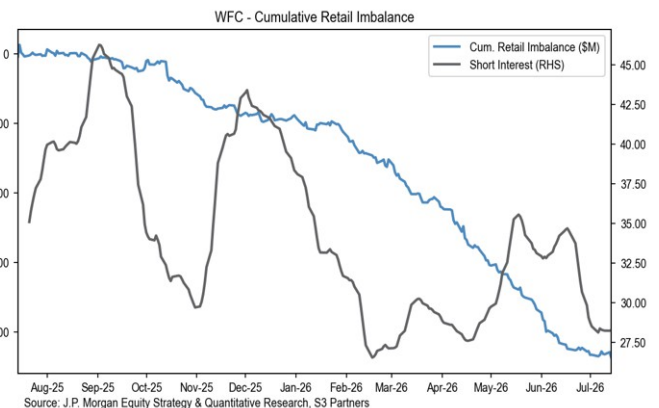
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 18: Cumulative Retail Imbalance in WFC

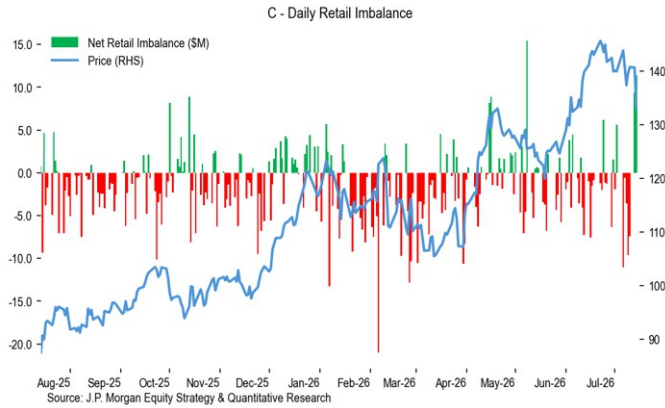
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 19: Retail Imbalance in C

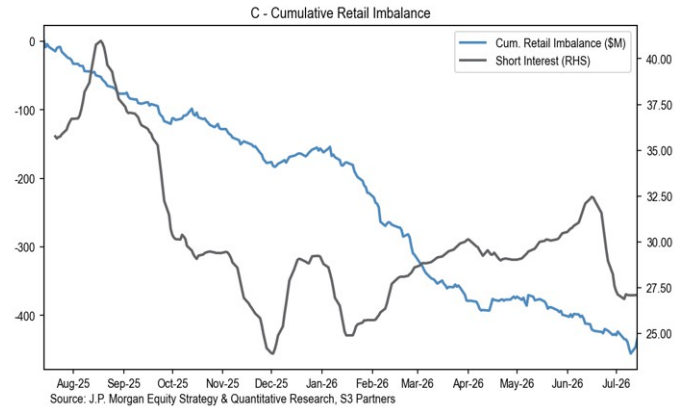
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 20: Cumulative Retail Imbalance in C

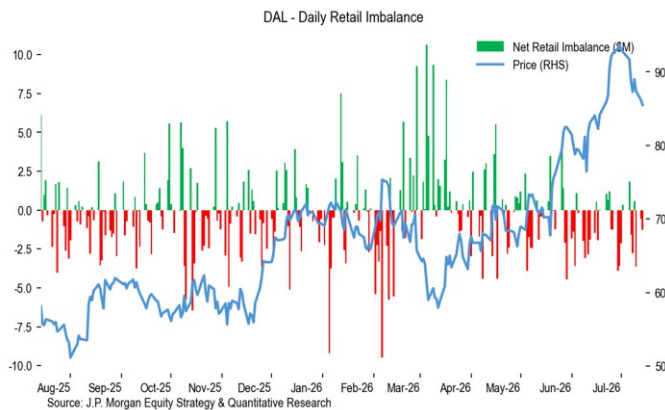
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 21: Retail Imbalance in DAL

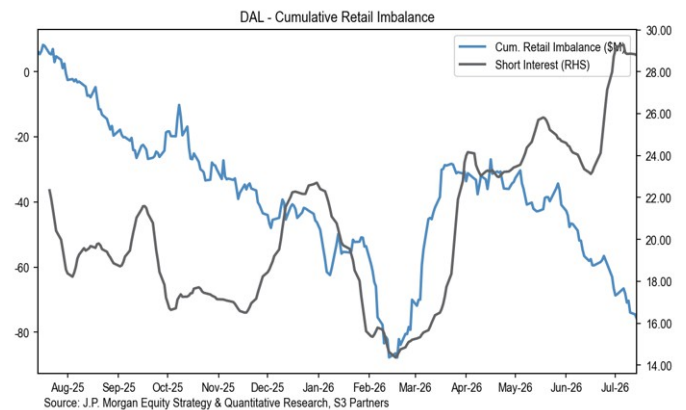
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 22: Cumulative Retail Imbalance in DAL

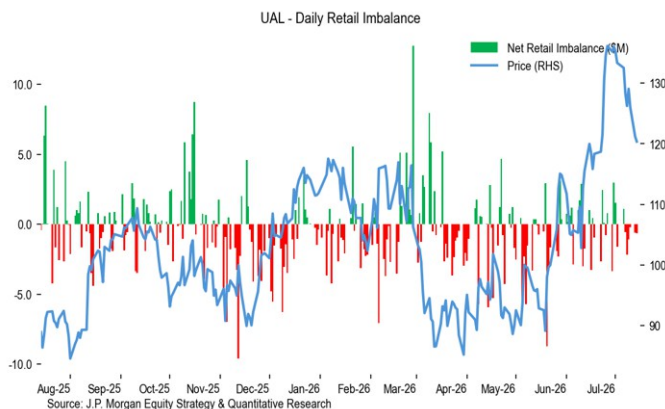
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 23: Retail Imbalance in UAL

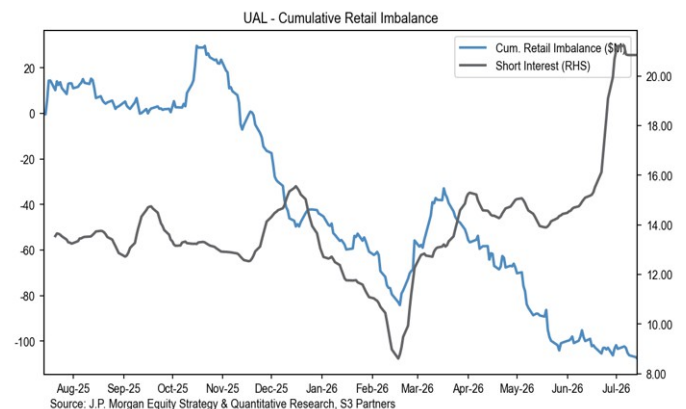
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 24: Cumulative Retail Imbalance in UAL

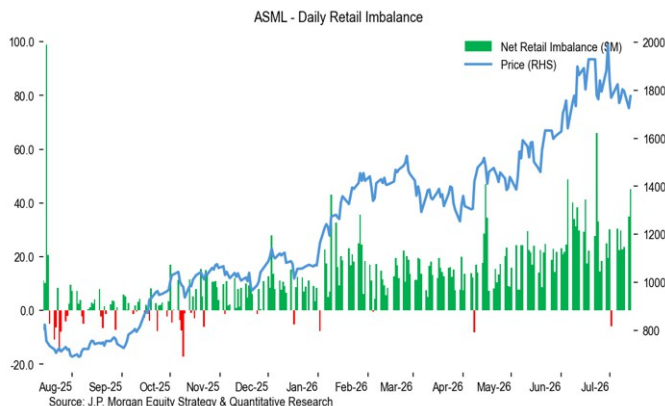
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 25: Retail Imbalance in ASML

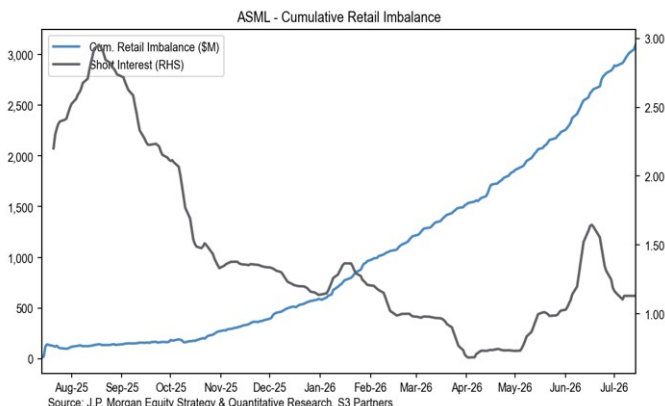
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 26: Cumulative Retail Imbalance in ASML

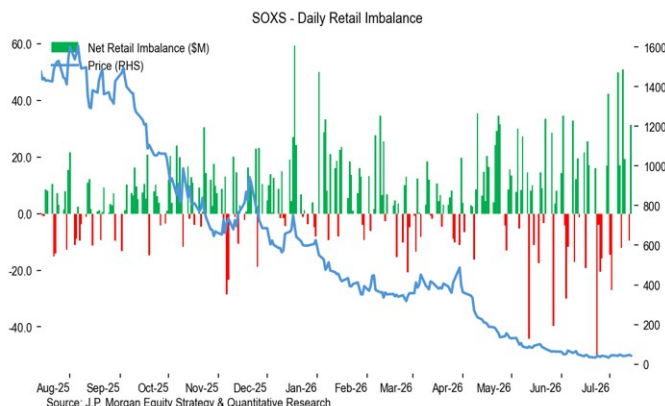
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 27: Retail Imbalance in SOXS

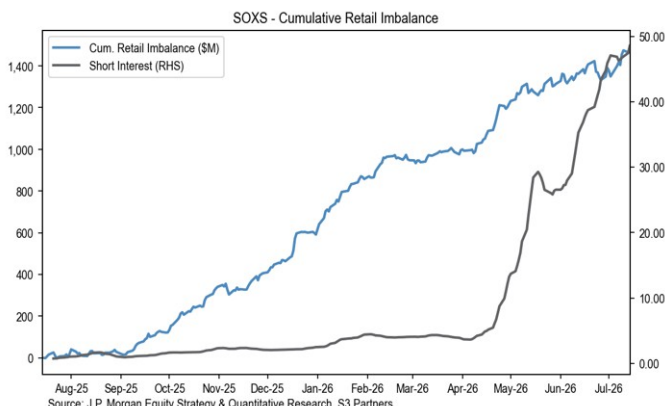
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 28: Cumulative Retail Imbalance in SOXS

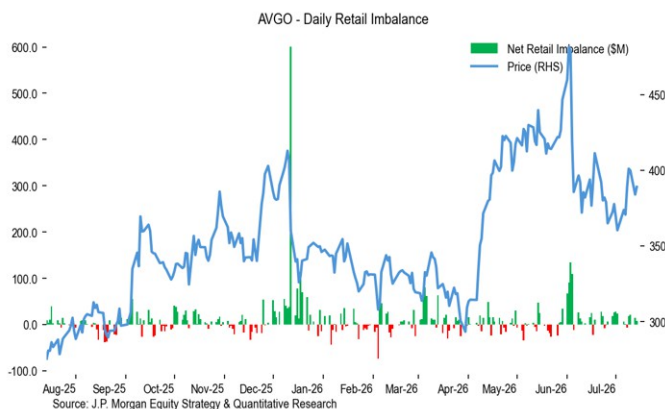
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 29: Retail Imbalance in AVGO

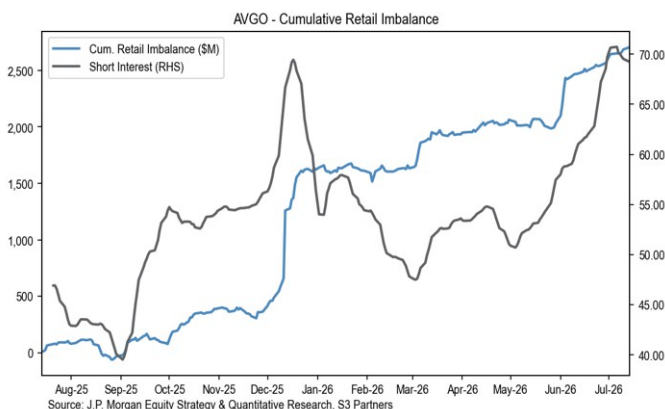
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 30: Cumulative Retail Imbalance in AVGO

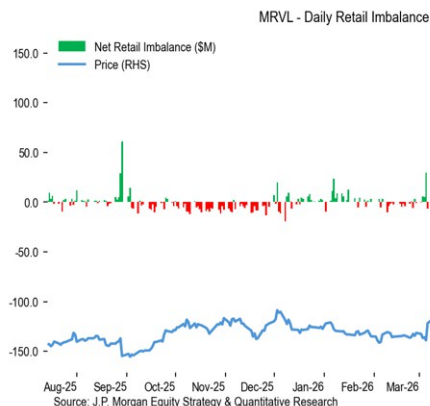
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 31: Retail Imbalance in MRVL

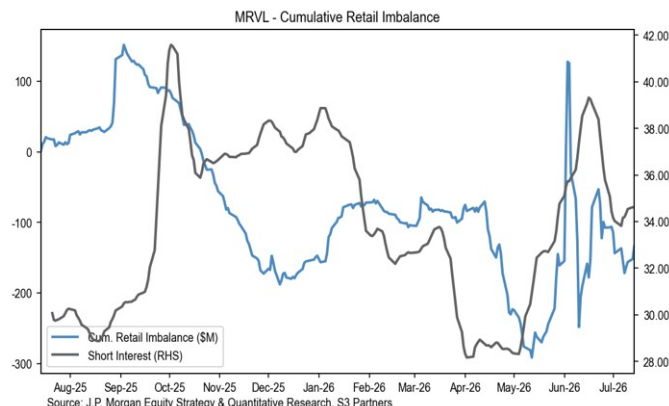
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 32: Cumulative Retail Imbalance in MRVL

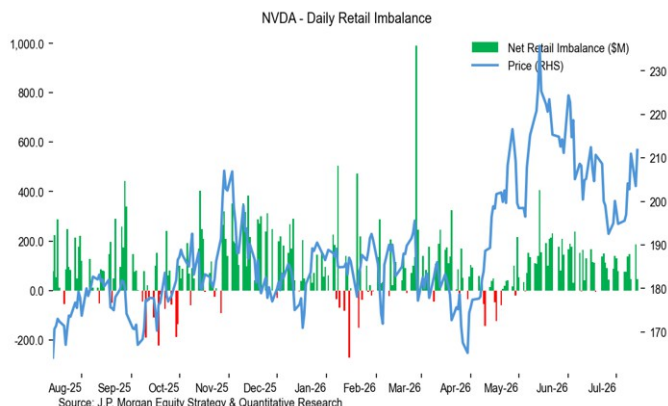
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 33: Retail Imbalance in NVDA

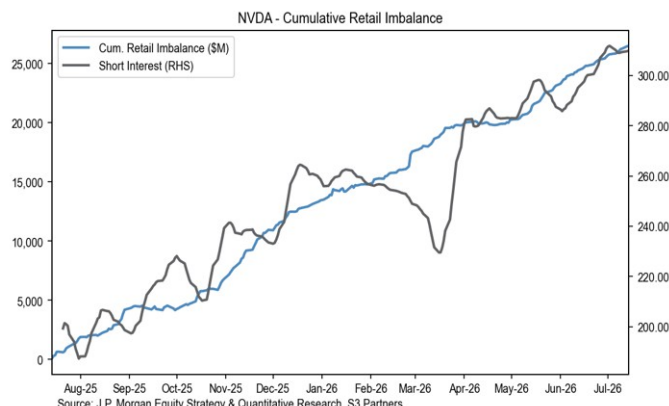
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 34: Cumulative Retail Imbalance in NVDA

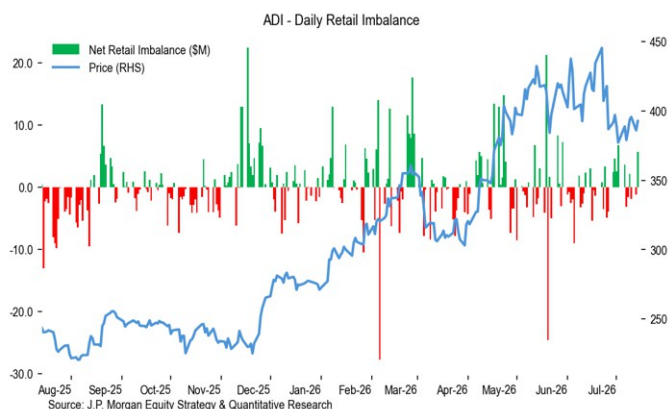
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 35: Retail Imbalance in ADI

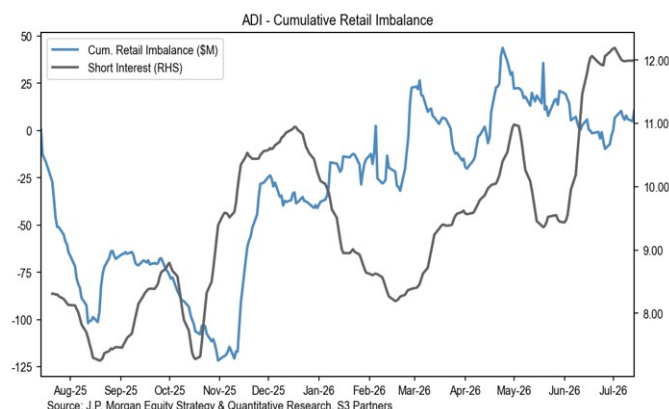
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 36: Cumulative Retail Imbalance in ADI

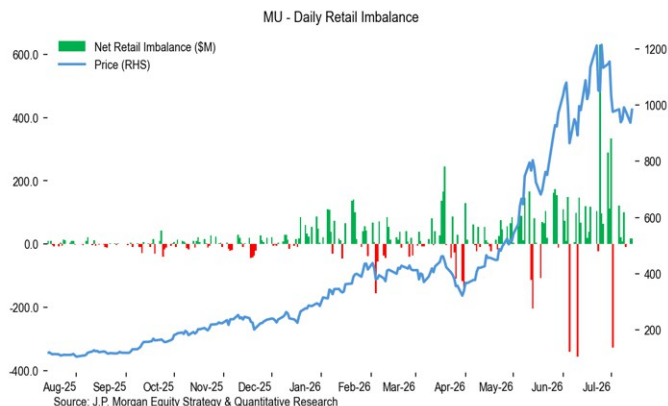
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 37: Retail Imbalance in MU

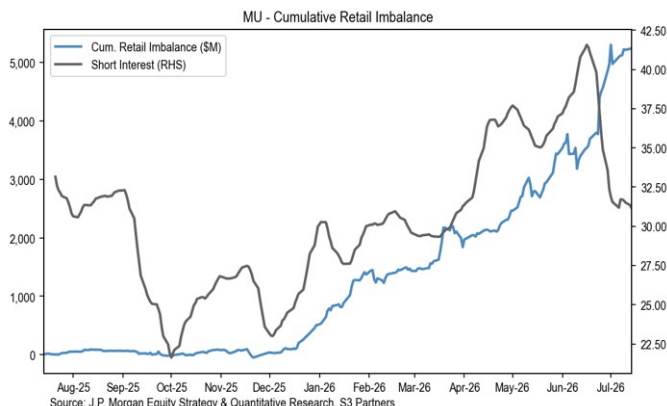
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 38: Cumulative Retail Imbalance in MU

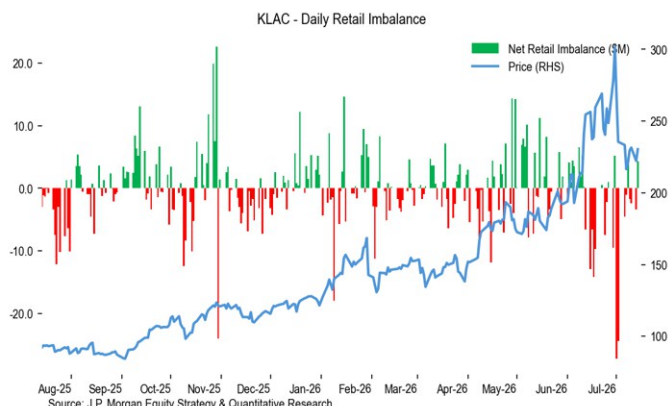
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 39: Retail Imbalance in KLAC

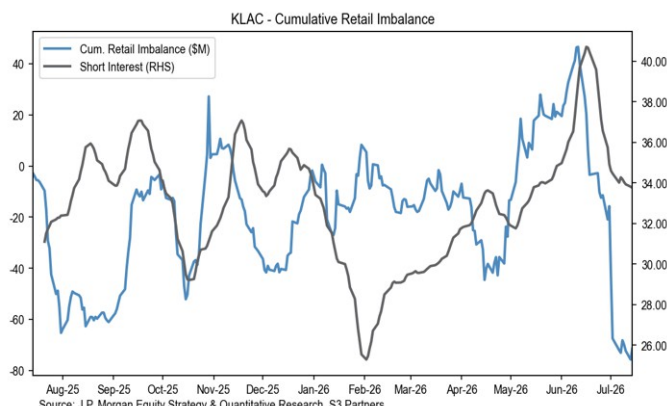
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 40: Cumulative Retail Imbalance in KLAC

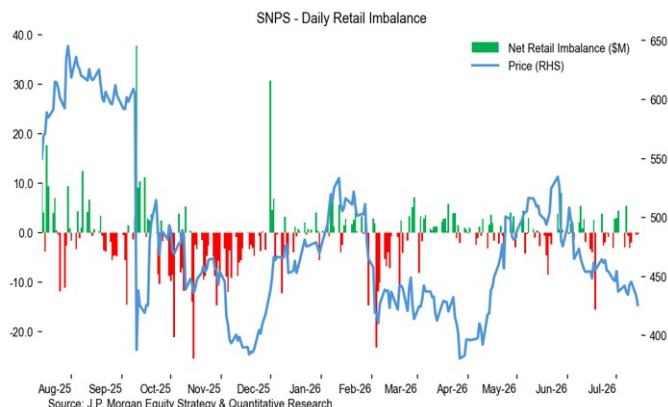
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 41: Retail Imbalance in SNPS

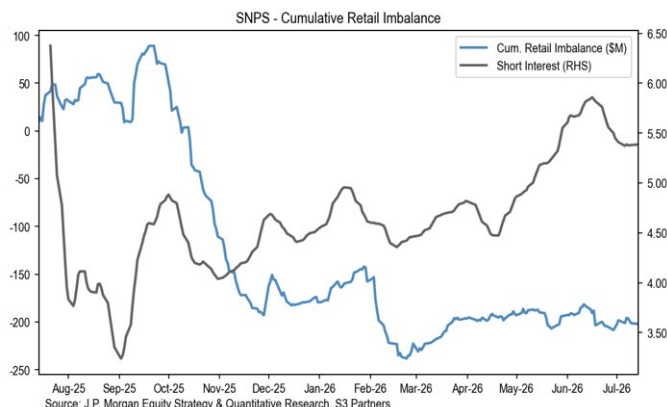
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 42: Cumulative Retail Imbalance in SNPS

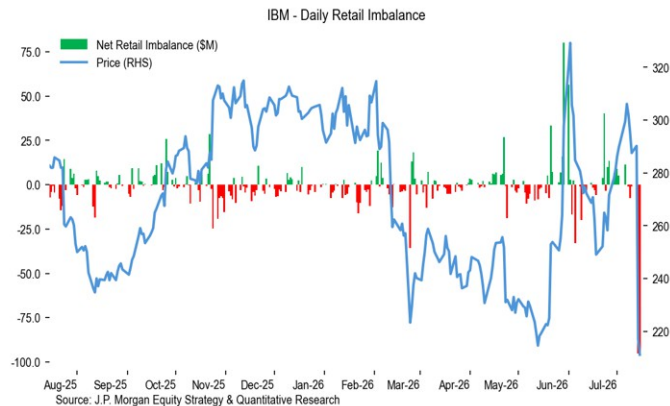
As of July 14th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 43: Retail Imbalance in IBM

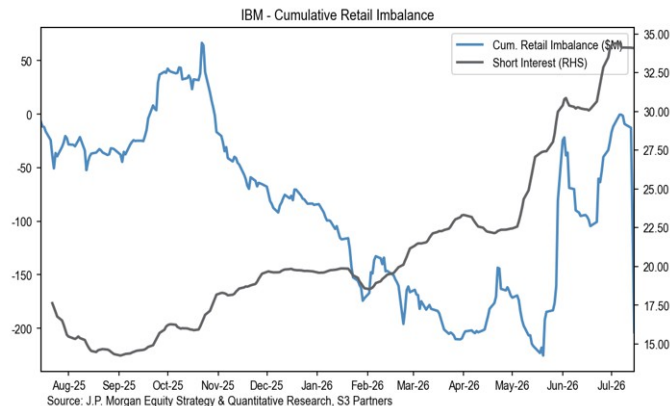
As of July 15th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

Figure 44: Cumulative Retail Imbalance in IBM

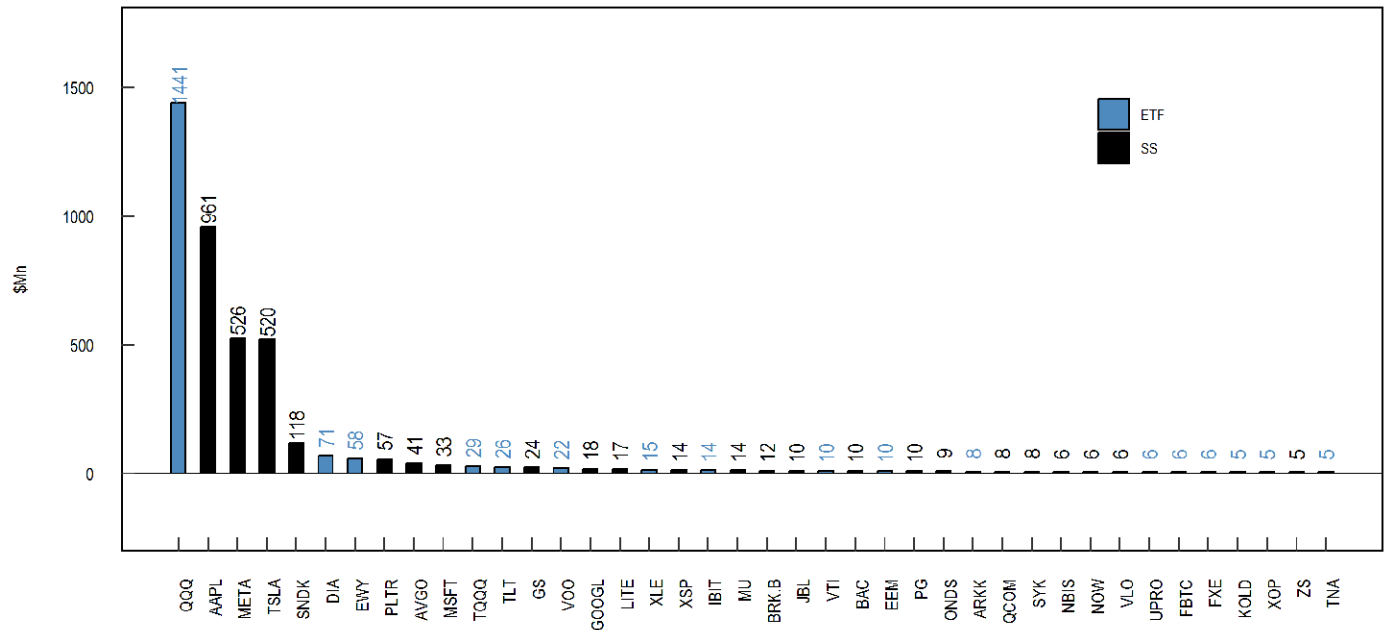
As of July 15th



Source: J.P. Morgan Equity Strategy & Quantitative Research, S3 Partners

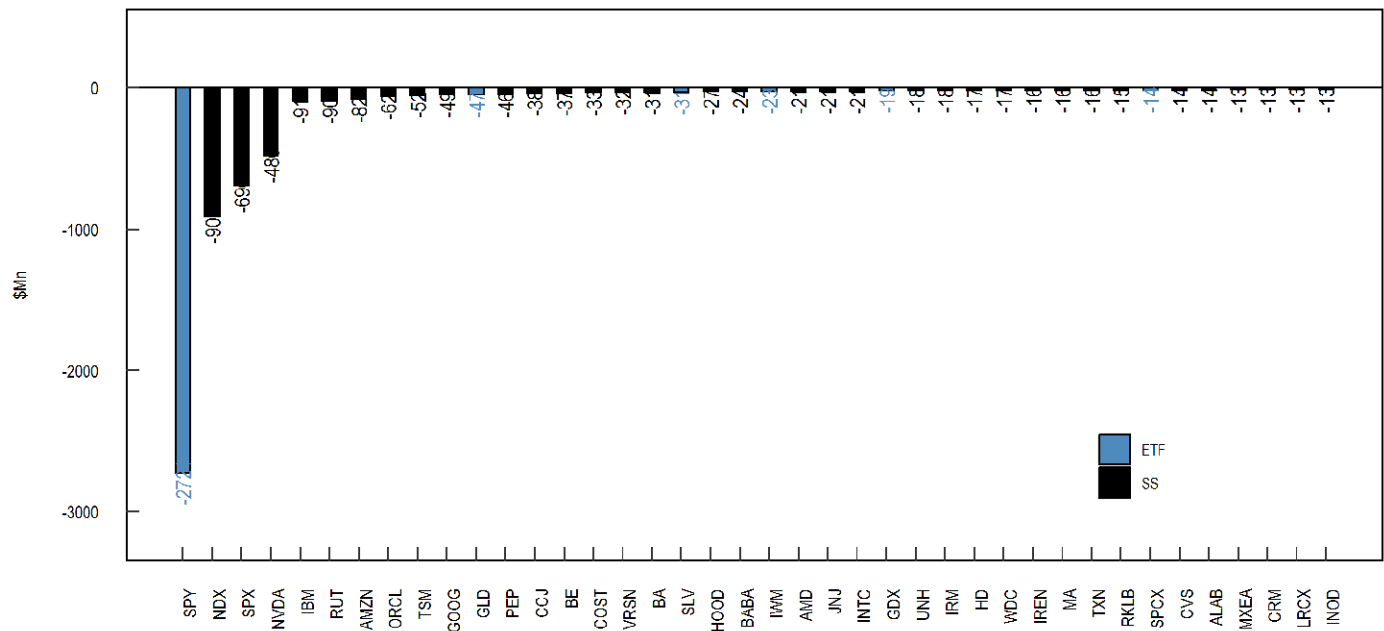
Retail Activity in Options

Figure 45: Top 40 Equities with Most Delta Bought (in \$Mn)



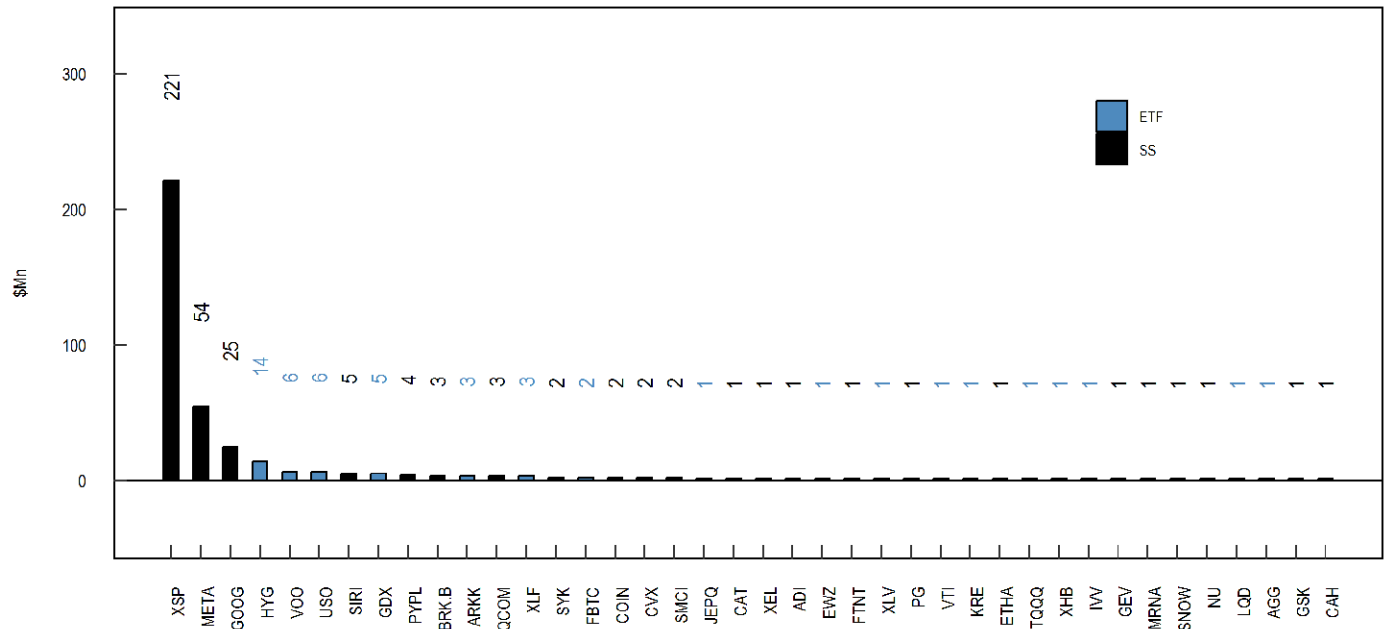
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 46: Top 40 Equities with Most Delta Sold (in \$Mn)



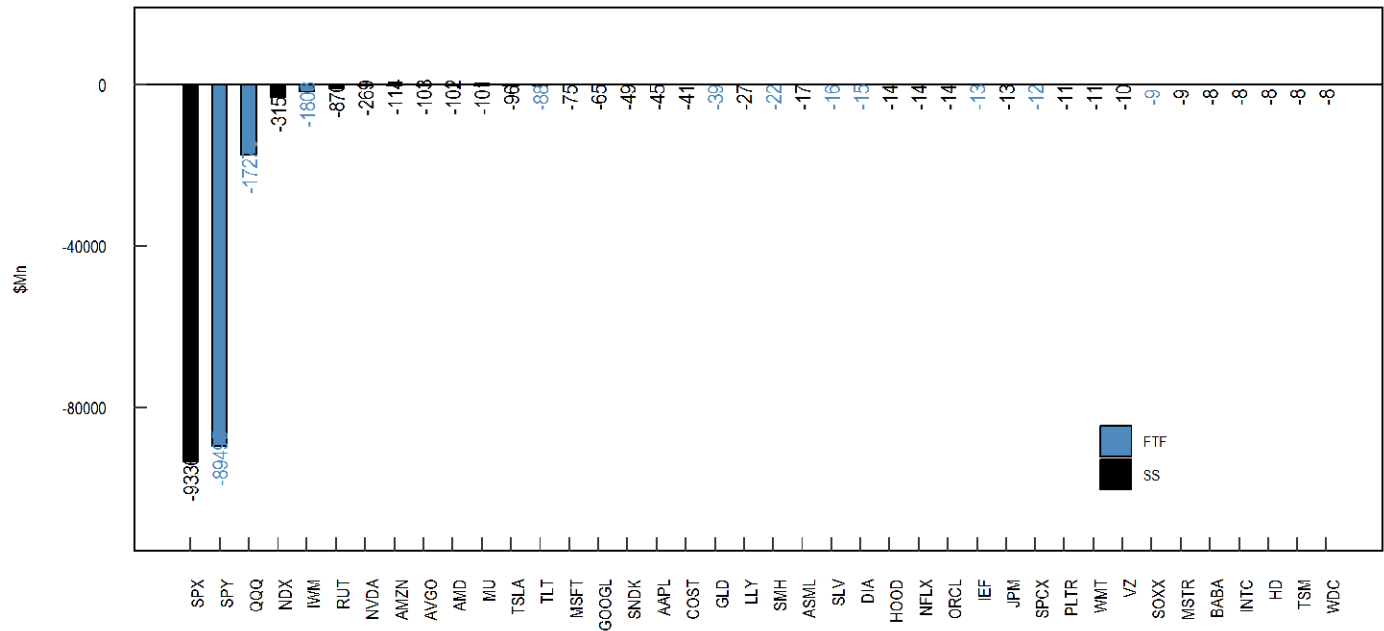
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 47: Top 40 Equities with Most Gamma Bought (in \$Mn)



Source: J.P. Morgan Equity Strategy & Quantitative Research

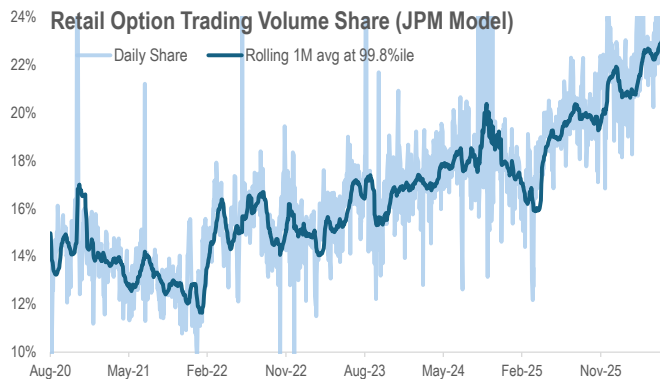
Figure 48: Top 40 Equities with Most Gamma Sold (in \$Mn)



Source: J.P. Morgan Equity Strategy & Quantitative Research

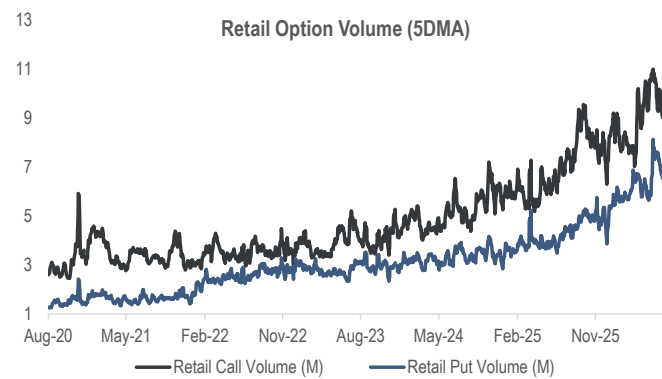
Retail Investors Options Activity by Sectors

Figure 49: Retail Options Trading Share



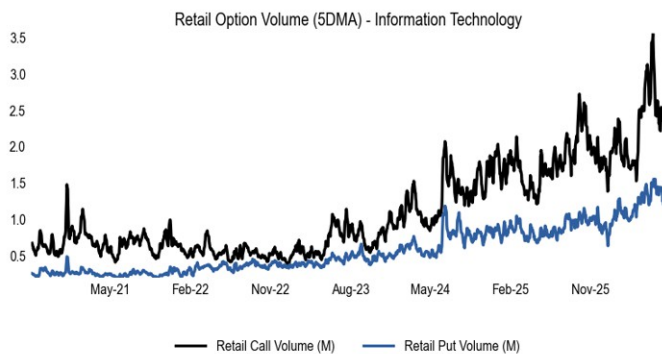
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 50: Retail Options Volume (Calls and Puts)



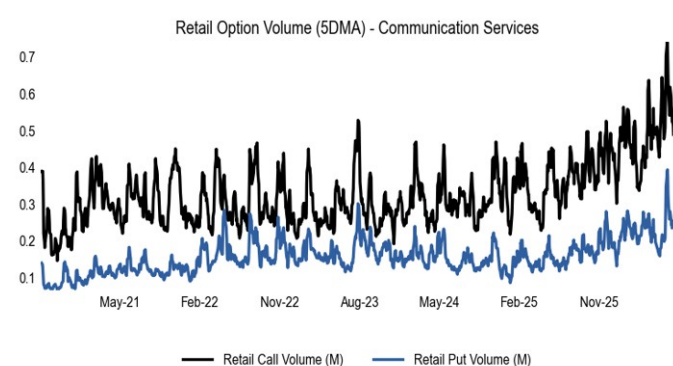
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 51: Retail Options Volume (Calls and Puts), Information Technology



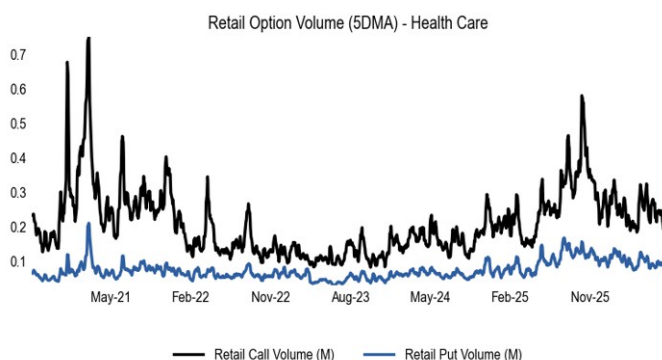
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 52: Retail Options Volume (Calls and Puts), Communication Services



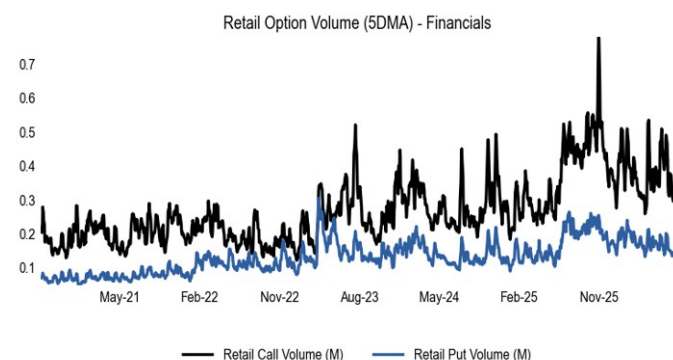
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 53: Retail Options Volume (Calls and Puts), Health Care



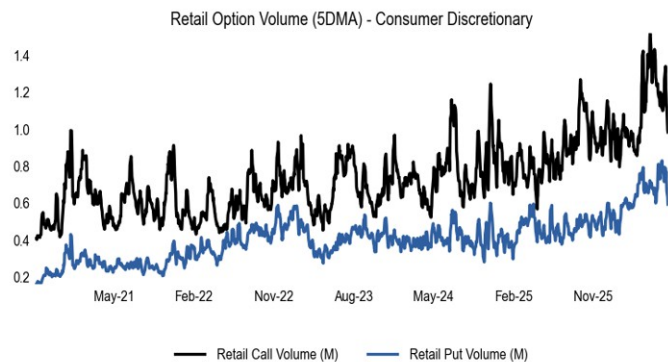
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 54: Retail Options Volume (Calls and Puts), Financials



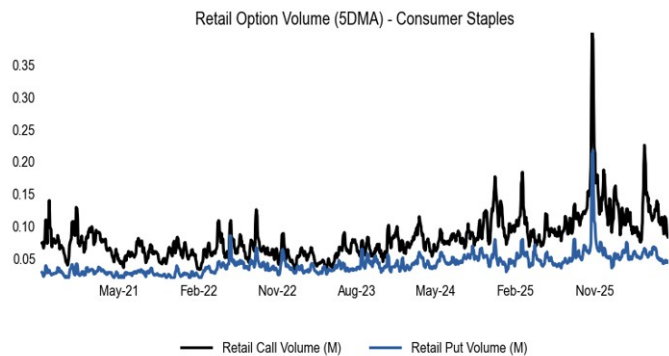
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 55: Retail Options Volume (Calls and Puts), Discretionary



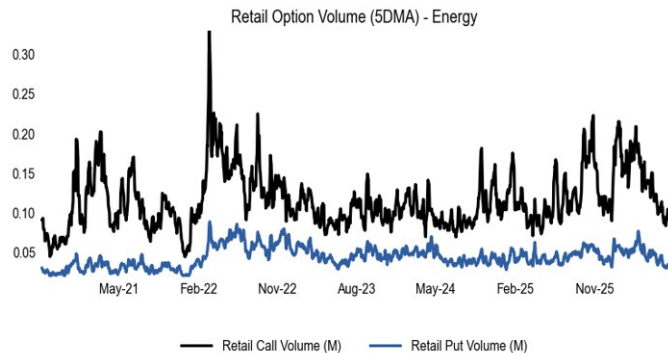
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 56: Retail Options Volume (Calls and Puts), Staples



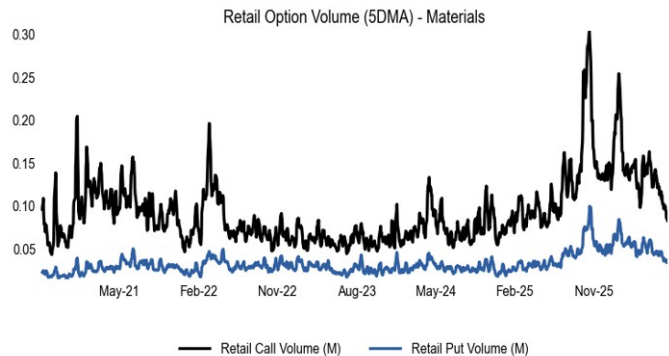
Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 57: Retail Options Volume (Calls and Puts), Energy



Source: J.P. Morgan Equity Strategy & Quantitative Research

Figure 58: Retail Options Volume (Calls and Puts), Materials



Source: J.P. Morgan Equity Strategy & Quantitative Research

Appendix: Alternative Data Points on Retail Activity

Figure 59: Retail Brokerage Volume Hit Record ~37% in January 2021

Source Flow by Retail Brokerage, % of Overall US Equity Mkt Volume (Stock + ETF)

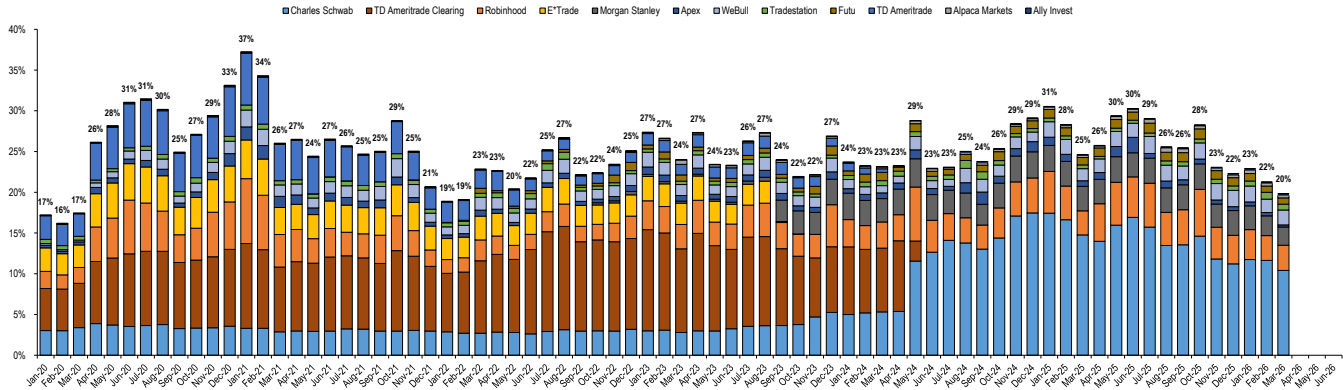
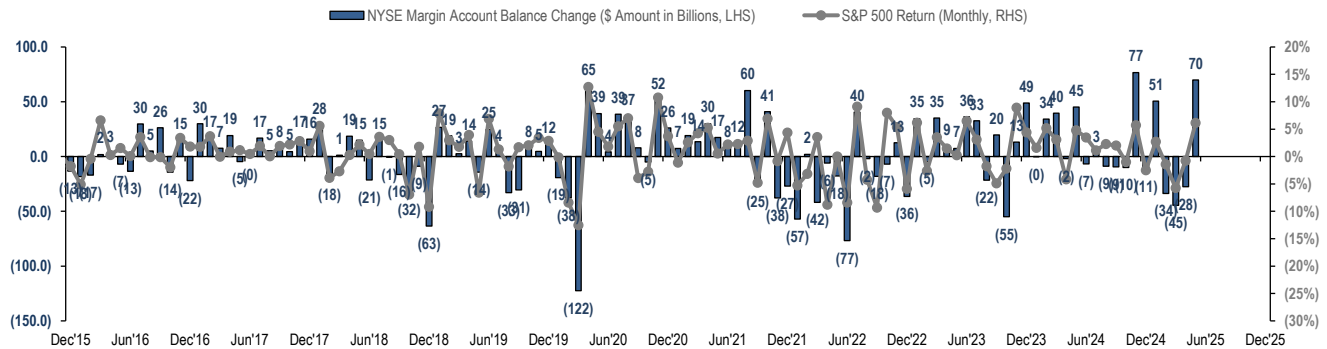


Figure 60: NYSE Margin Account – Monthly Net Debit Balance Change

\$ in Billions, Net Debit balance equals margin debit balance less total credit balances



Source: J.P. Morgan Equity Strategy & Quantitative Research, Bloomberg Finance L.P., FINRA, Quarterly 606 Reports (for two figures above).

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/performance in line/underperform the average total return of the stocks in the Research Analyst’s, or the Research Analyst’s team’s, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock’s expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts’ coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst’s coverage universe can be found on J.P. Morgan’s Research website, <https://>.

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia’s Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE (“JPM SE”), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions (“EEA professional investors”). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a ‘Research Analyst’ having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpimpl.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpimpl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpimpl.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 15 Jul 2026 09:47 PM EDT

Disseminated 15 Jul 2026 09:47 PM EDT